

$$\beta = (X^{\top} X + \lambda I)^{-1} X^{\top} y$$

compared with the normal equation, only
 $+\lambda I$ is new; it adds λ to the diagonal

$$I = \begin{bmatrix} \mathbf{0} & 0 & 0 & \cdots & 0 \\ 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & & & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \end{bmatrix}$$

top-left entry 0: the
intercept β_0 is not penalised